

Sino Cicatrizado (The Scarred Bell)

<https://map.sympoietic.systems>

It turns Ouro Preto into a living, mutating soundscape. Your phone broadcasts your position. The city replies with bells. Every interaction leaves a trace that the next person hears.

The core cybernetic move is simple: we put the observer inside the loop. You are not looking at a map from above, immune and detached. You are a somatic node — your body co-constitutes the system. No position of objectivity. No undo.

Here's the three things that drive it.

The bells of Ouro Preto.

For centuries, before the telegraph, these church bells functioned as a city-wide information network. Different tolls signaled births, deaths, fires, invasions — by demographic. Man, woman, child, priest. The enslaved *sineiros*, the bell ringers, were forced to perform rigid European ecclesiastical scores. But from up in those towers, they injected African syncopation and polyrhythms — hidden messages of safety and rebellion embedded in the colonial soundscape. That's acoustic resistance.

Our virtual bells inherit that lineage. The underlying rhythms are Euclidean, mathematically precise. But as you approach a tower, your presence detunes the frequencies and injects syncopation. The algorithm gets reclaimed by a body walking past it.

Nego Bispo.

He was a Piaui intellectual, rooted in quilombola cosmology, who drew a crucial distinction — *transfluence* versus *confluence*. Transfluence is the colonial move: rivers merge, differences get erased, the stronger current assimilates the weaker. Confluence is currents meeting, mutually amplifying, each sediment trail staying visible. That's how our system handles two participants crossing paths — their acoustic signatures mix, but neither gets flattened into the other.

And this also echoes Chico Rei — the Kongo king enslaved in these very Ouro Preto mines. He covertly accumulated gold dust in his hair, washed it out at night, bought his own freedom, then his son's, then his entire community's. Eventually he bought the mine itself — the Encardideira. He subverted the apparatus from within. The Church of Santa Efigênia, funded by that emancipated community, still stands.

The environment as co-composer.

Drawing on Mark Hansen's critique of autopoietic closure — the idea that a system can be structurally closed while still coupled to its environment — we treat the city as an active agent. GPS drift doesn't get smoothed — it fluctuates the rhythm generators, creating micro-timing syncopations. Cellular packet loss routes straight into a bit-crusher, producing digital grit. As your phone battery drains below fifty percent, the audio degrades from sixteen-bit clarity down to four-bit. The hardware performs its own exhaustion. This is a System-Environment Hybrid: the boundary between system and world is porous by design.

Here's what you actually do.

1. You walk Ouro Preto with the site open. Your phone becomes what we call a somatic node — your position broadcasts live into the system.
2. You can trigger a echo: an acoustic probe that travels outward at the physical speed of sound — three hundred forty-three meters per second — returning delayed echoes from church towers and other participants' reflectors up to five hundred meters away. You hear the scarred history of the space in real time. This is the witness-function of second-order cybernetics made acoustic — you read-listen to the invisible somatic density around you. The ghosts reply.
3. You can drop a reflector at your location — a permanent marker. It carries an acoustic preset generated from your intent, your words, run through an LLM membrane into a JSON patch. That reflector stays. It becomes part of what the next person hears.